IB Economics - Topic 2.8: Teacher and AI Grading Guide

Market Failure: Externalities and Common-Pool Resources | Questions 1-50

Scope and assessment status

Each original paired question has (a) 10 marks and (b) 15 marks. Questions 1-45 are SL/HL. Questions 46-50 include HL-only welfare-loss calculations. Difficulty: 10 Easy, 25 Medium and 15 Hard pairs, distributed through the bank. Level and difficulty are independent.

Current Topic 2.8 covers MSB=MSC social optimum; four externality types, welfare loss, merit/demerit goods, rivalrous non-excludable common-pool resources and sustainability; taxes, permits, regulation, education, subsidies, provision, collective governance, international cooperation; policy strengths/limits. Diagram-based welfare-loss calculation is HL only.

Uploaded older Economics for the IB Diploma Chapter 5 covers externalities and common-access resources. It is supplementary; current syllabus lists the specific response instruments and HL calculation boundary.

These are original adapted Paper 1-style practice questions, not official IB examination or marking material. The question bank revisits syllabus mechanisms through distinct explanatory and evaluative tasks; the pairs are alternatives for practice, not separate official learning outcomes.

Coverage blueprint

Social optimum and externality definitions: 1-5

Negative production: 6-10

Negative consumption: 11-15

Positive production: 16-20

Positive consumption: 21-25

Common-pool resources: 26-30

Policy options and evaluation: 31-40

International cooperation: 41-45

HL welfare-loss calculations: 46-50

Marking principles

Use best-fit assessment for each part, not a fixed marks-per-keyword checklist. Credit accurate alternatives addressing the task. Do not require every example or qualification below.

Part (a): 10 marks

Practice bands: 0; 1-2 limited; 3-4 mainly descriptive; 5-6 partial explanation; 7-8 clear and accurate causal analysis; 9-10 fully developed explanation with any required diagram or calculation. These abbreviated bands are not official IB descriptors.

Part (b): 15 marks

Practice bands: 0; 1-3 very limited; 4-6 superficial; 7-9 developed but weakly balanced; 10-12 balanced analysis and evaluation; 13-15 sustained, applied evaluation and supported judgment. A named case alone is not developed application.

Diagram and AI grading

For each externality, first identify whether the wedge is between MSC/MPC or MSB/MPB, then mark Qm at private intersection and Q* at MSB=MSC. Negative externalities generally Qm>Q*, positive Qm<Q*. Welfare-loss area consists of marginal social cost-benefit gaps over the inefficient units. No numeric loss is required for SL/HL questions; HL numerical examples specify linearity.

Record separate marks for (a) and (b), band evidence, diagram or numerical features if relevant, economic misconceptions, quality of real-world application, and one actionable improvement per part. If a student drawing is unclear, flag the uncertainty rather than assuming correctness.

Question 1

Easy | SL/HL | 25 marks

Topic 2.8: Social optimum

(a) [10] Explain why MSB=MSC identifies an allocatively efficient quantity.

(b) [15] Using real-world examples, discuss whether market equilibrium always coincides with the social optimum.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Equality means no additional unit creates net social gain, assuming all effects counted.

Expected diagram or calculation. P or marginal value vertically and Q horizontally; MSB=MSC at Q*, explain net benefits either side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes social optimum from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Equality means no additional unit creates net social gain, assuming all effects counted.

Counterargument and conditions. Market valuation may omit third-party costs/benefits and estimates are imperfect.

Application and real-world example. A named pollution or vaccination case with documented context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 2

Medium | SL/HL | 25 marks

Topic 2.8: Social optimum

(a) [10] Using an appropriate diagram, explain why output below the social optimum leaves possible gains unrealised.

(b) [15] Using real-world examples, evaluate whether higher output always increases social welfare.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. For Q below MSB=MSC, MSB>MSC, so additional units have positive net benefit.

Expected diagram or calculation. P or marginal value vertically and Q horizontally; MSB=MSC at Q*, explain net benefits either side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes social optimum from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. For Q below MSB=MSC, MSB>MSC, so additional units have positive net benefit.

Counterargument and conditions. Market valuation may omit third-party costs/benefits and estimates are imperfect.

Application and real-world example. A named pollution or vaccination case with documented context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 3

Hard | SL/HL | 25 marks

Topic 2.8: Social optimum

(a) [10] Using an appropriate diagram, explain why output above the social optimum reduces community surplus.

(b) [15] Using real-world examples, discuss whether maximum physical output should be a policy aim.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Beyond Q*, MSC>MSB, so extra units cost society more than they benefit.

Expected diagram or calculation. P or marginal value vertically and Q horizontally; MSB=MSC at Q*, explain net benefits either side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes social optimum from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Beyond Q*, MSC>MSB, so extra units cost society more than they benefit.

Counterargument and conditions. Market valuation may omit third-party costs/benefits and estimates are imperfect.

Application and real-world example. A named pollution or vaccination case with documented context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 4

Medium | SL/HL | 25 marks

Topic 2.8: Social optimum

(a) [10] Explain the difference between a marginal private and a marginal social cost.

(b) [15] Using real-world examples, evaluate whether producers normally observe all costs their output imposes.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSC includes external cost not borne by decision-maker.

Expected diagram or calculation. P or marginal value vertically and Q horizontally; MSB=MSC at Q*, explain net benefits either side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes social optimum from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSC includes external cost not borne by decision-maker.

Counterargument and conditions. Market valuation may omit third-party costs/benefits and estimates are imperfect.

Application and real-world example. A named pollution or vaccination case with documented context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 5

Medium | SL/HL | 25 marks

Topic 2.8: Social optimum

(a) [10] Explain the difference between a marginal private and marginal social benefit.

(b) [15] Using real-world examples, discuss whether private willingness to pay fully measures social benefit.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSB includes spillover benefit not captured privately.

Expected diagram or calculation. P or marginal value vertically and Q horizontally; MSB=MSC at Q*, explain net benefits either side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes social optimum from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSB includes spillover benefit not captured privately.

Counterargument and conditions. Market valuation may omit third-party costs/benefits and estimates are imperfect.

Application and real-world example. A named pollution or vaccination case with documented context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 6

Easy | SL/HL | 25 marks

Topic 2.8: Negative production

(a) [10] Using an appropriate diagram, explain overproduction arising from a negative externality of production.

(b) [15] Using real-world examples, evaluate whether polluting firms have an incentive to reduce output without intervention.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSC above MPC, market Q where MPB=MPC exceeds social Q where MSB=MSC.

Expected diagram or calculation. P/MB/MC vertical, Q horizontal; MSC above MPC, MSB=MPB, Qm>Q*, shade welfare-loss triangle between Q* and Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSC above MPC, market Q where MPB=MPC exceeds social Q where MSB=MSC.

Counterargument and conditions. Damage measurement and alternative policies matter; an observed output change does not alone prove efficiency.

Application and real-world example. World Bank carbon pricing context for industrial emissions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 7

Hard | SL/HL | 25 marks

Topic 2.8: Negative production

(a) [10] Explain why air pollution from a factory is a production externality.

(b) [15] Using real-world examples, discuss whether a pollution tax can fully internalise the external cost.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Third parties bear pollution cost; appropriate per-unit tax raises private cost toward MSC.

Expected diagram or calculation. P/MB/MC vertical, Q horizontal; MSC above MPC, MSB=MPB, Qm>Q*, shade welfare-loss triangle between Q* and Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Third parties bear pollution cost; appropriate per-unit tax raises private cost toward MSC.

Counterargument and conditions. Damage measurement and alternative policies matter; an observed output change does not alone prove efficiency.

Application and real-world example. World Bank carbon pricing context for industrial emissions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 8

Medium | SL/HL | 25 marks

Topic 2.8: Negative production

(a) [10] Using an appropriate diagram, explain the welfare loss from a negative production externality.

(b) [15] Using real-world examples, evaluate whether measuring environmental damage is necessary for an optimal tax.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Between social and market quantities MSC exceeds MSB, generating a triangular net loss.

Expected diagram or calculation. P/MB/MC vertical, Q horizontal; MSC above MPC, MSB=MPB, Qm>Q*, shade welfare-loss triangle between Q* and Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Between social and market quantities MSC exceeds MSB, generating a triangular net loss.

Counterargument and conditions. Damage measurement and alternative policies matter; an observed output change does not alone prove efficiency.

Application and real-world example. World Bank carbon pricing context for industrial emissions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 9

Hard | SL/HL | 25 marks

Topic 2.8: Negative production

(a) [10] Explain why a firm's private production cost may be lower than the social cost.

(b) [15] Using real-world examples, discuss whether voluntary reductions in pollution are sufficient to correct market failure.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Unpaid harm to others is excluded from MPC.

Expected diagram or calculation. P/MB/MC vertical, Q horizontal; MSC above MPC, MSB=MPB, Qm>Q*, shade welfare-loss triangle between Q* and Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Unpaid harm to others is excluded from MPC.

Counterargument and conditions. Damage measurement and alternative policies matter; an observed output change does not alone prove efficiency.

Application and real-world example. World Bank carbon pricing context for industrial emissions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 10

Medium | SL/HL | 25 marks

Topic 2.8: Negative production

(a) [10] Using an appropriate diagram, explain how an indirect tax could reduce output toward the social optimum in a polluting industry.

(b) [15] Using real-world examples, evaluate the effectiveness of a carbon tax.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Tax shifts private supply toward MSC, reducing excessive Q if correctly set.

Expected diagram or calculation. P/MB/MC vertical, Q horizontal; MSC above MPC, MSB=MPB, Qm>Q*, shade welfare-loss triangle between Q* and Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Tax shifts private supply toward MSC, reducing excessive Q if correctly set.

Counterargument and conditions. Damage measurement and alternative policies matter; an observed output change does not alone prove efficiency.

Application and real-world example. World Bank carbon pricing context for industrial emissions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 11

Easy | SL/HL | 25 marks

Topic 2.8: Negative consumption

(a) [10] Using an appropriate diagram, explain market failure from a negative externality of consumption.

(b) [15] Using real-world examples, discuss whether discouraging consumption is enough to remove third-party harms.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSB below MPB, so market Q exceeds efficient Q under uncorrected decision.

Expected diagram or calculation. MSB below MPB, MPC=MSC, Qm>Q*, welfare loss between MSC and MSB.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSB below MPB, so market Q exceeds efficient Q under uncorrected decision.

Counterargument and conditions. External harm varies and price measures may be incomplete.

Application and real-world example. Second-hand smoke in a named jurisdiction, with documented law.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 12

Medium | SL/HL | 25 marks

Topic 2.8: Negative consumption

(a) [10] Explain how second-hand smoke can create a negative externality of consumption.

(b) [15] Using real-world examples, evaluate whether regulation is preferable to taxation in this context.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Non-smokers incur uncompensated harm from consumption.

Expected diagram or calculation. MSB below MPB, MPC=MSC, Qm>Q*, welfare loss between MSC and MSB.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Non-smokers incur uncompensated harm from consumption.

Counterargument and conditions. External harm varies and price measures may be incomplete.

Application and real-world example. Second-hand smoke in a named jurisdiction, with documented law.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 13

Hard | SL/HL | 25 marks

Topic 2.8: Negative consumption

(a) [10] Using an appropriate diagram, explain why private willingness to pay can exceed social benefit for a demerit good.

(b) [15] Using real-world examples, discuss whether all consumers undervalue health risks of demerit goods.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Demerit good may be overconsumed due to external harm or imperfect appreciation; do not assume every buyer ignorant.

Expected diagram or calculation. MSB below MPB, MPC=MSC, Qm>Q*, welfare loss between MSC and MSB.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Demerit good may be overconsumed due to external harm or imperfect appreciation; do not assume every buyer ignorant.

Counterargument and conditions. External harm varies and price measures may be incomplete.

Application and real-world example. Second-hand smoke in a named jurisdiction, with documented law.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 14

Medium | SL/HL | 25 marks

Topic 2.8: Negative consumption

(a) [10] Explain why a tax on a demerit good may affect both consumers and producers.

(b) [15] Using real-world examples, evaluate whether indirect taxes are an equitable response to demerit goods.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Higher buyer price reduces Q while seller price/revenue can change.

Expected diagram or calculation. MSB below MPB, MPC=MSC, Qm>Q*, welfare loss between MSC and MSB.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Higher buyer price reduces Q while seller price/revenue can change.

Counterargument and conditions. External harm varies and price measures may be incomplete.

Application and real-world example. Second-hand smoke in a named jurisdiction, with documented law.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 15

Medium | SL/HL | 25 marks

Topic 2.8: Negative consumption

(a) [10] Using an appropriate diagram, explain why a lower market price does not necessarily indicate efficient consumption of a harmful good.

(b) [15] Using real-world examples, discuss whether price information alone captures social costs.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Private demand ignores third-party harm, leading to Qmarket>Qsocial.

Expected diagram or calculation. MSB below MPB, MPC=MSC, Qm>Q*, welfare loss between MSC and MSB.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes negative consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Private demand ignores third-party harm, leading to Qmarket>Qsocial.

Counterargument and conditions. External harm varies and price measures may be incomplete.

Application and real-world example. Second-hand smoke in a named jurisdiction, with documented law.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 16

Easy | SL/HL | 25 marks

Topic 2.8: Positive production

(a) [10] Using an appropriate diagram, explain underproduction caused by a positive production externality.

(b) [15] Using real-world examples, evaluate whether firms can capture enough benefits from research to supply the social optimum.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSC below MPC, so private supply yields Qmarket<Qsocial.

Expected diagram or calculation. MSC below MPC, MSB=MPB, Qm<Q*, forgone benefit triangle.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSC below MPC, so private supply yields Qmarket<Qsocial.

Counterargument and conditions. Spillovers vary, and firms may capture part of benefit.

Application and real-world example. Research spillovers in a named sector if documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 17

Hard | SL/HL | 25 marks

Topic 2.8: Positive production

(a) [10] Explain how a producer's training may benefit other firms.

(b) [15] Using real-world examples, discuss whether subsidies for training always increase social welfare.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Skills can spill over when workers move, creating external production benefit.

Expected diagram or calculation. MSC below MPC, MSB=MPB, Qm<Q*, forgone benefit triangle.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Skills can spill over when workers move, creating external production benefit.

Counterargument and conditions. Spillovers vary, and firms may capture part of benefit.

Application and real-world example. Research spillovers in a named sector if documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 18

Medium | SL/HL | 25 marks

Topic 2.8: Positive production

(a) [10] Using an appropriate diagram, explain how a subsidy could increase output toward the social optimum when production has external benefits.

(b) [15] Using real-world examples, evaluate the difficulty of setting an appropriate subsidy.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Payment lowers producer's effective cost toward MSC, raising Q.

Expected diagram or calculation. MSC below MPC, MSB=MPB, Qm<Q*, forgone benefit triangle.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Payment lowers producer's effective cost toward MSC, raising Q.

Counterargument and conditions. Spillovers vary, and firms may capture part of benefit.

Application and real-world example. Research spillovers in a named sector if documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 19

Hard | SL/HL | 25 marks

Topic 2.8: Positive production

(a) [10] Explain why a firm's investment can generate benefits not reflected in its revenue.

(b) [15] Using real-world examples, discuss whether all spillovers require government support.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Knowledge diffusion can benefit others without payment; private incentives may still be strong.

Expected diagram or calculation. MSC below MPC, MSB=MPB, Qm<Q*, forgone benefit triangle.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Knowledge diffusion can benefit others without payment; private incentives may still be strong.

Counterargument and conditions. Spillovers vary, and firms may capture part of benefit.

Application and real-world example. Research spillovers in a named sector if documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 20

Medium | SL/HL | 25 marks

Topic 2.8: Positive production

(a) [10] Using an appropriate diagram, explain the welfare loss from underproduction with a positive production externality.

(b) [15] Using real-world examples, evaluate whether public provision is preferable to producer subsidies.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Between market and social Q, MSB exceeds MSC, so forgone net benefits.

Expected diagram or calculation. MSC below MPC, MSB=MPB, Qm<Q*, forgone benefit triangle.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive production from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Between market and social Q, MSB exceeds MSC, so forgone net benefits.

Counterargument and conditions. Spillovers vary, and firms may capture part of benefit.

Application and real-world example. Research spillovers in a named sector if documented.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 21

Easy | SL/HL | 25 marks

Topic 2.8: Positive consumption

(a) [10] Using an appropriate diagram, explain underconsumption arising from a positive externality of consumption.

(b) [15] Using real-world examples, discuss whether education is likely to be underconsumed without intervention.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. MSB above MPB, so market Q below social optimum.

Expected diagram or calculation. MSB above MPB, MSC=MPC, Qm<Q*, mark missing social surplus.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. MSB above MPB, so market Q below social optimum.

Counterargument and conditions. Information, capacity and willingness differ; subsidy need not ensure uptake.

Application and real-world example. UK vaccination programs, with documented external benefit mechanism.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 22

Medium | SL/HL | 25 marks

Topic 2.8: Positive consumption

(a) [10] Explain how vaccination can create a benefit for people who do not receive it.

(b) [15] Using real-world examples, evaluate whether public information alone can increase vaccination to the social optimum.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Reduced transmission can benefit others; information addresses only one barrier.

Expected diagram or calculation. MSB above MPB, MSC=MPC, Qm<Q*, mark missing social surplus.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Reduced transmission can benefit others; information addresses only one barrier.

Counterargument and conditions. Information, capacity and willingness differ; subsidy need not ensure uptake.

Application and real-world example. UK vaccination programs, with documented external benefit mechanism.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 23

Hard | SL/HL | 25 marks

Topic 2.8: Positive consumption

(a) [10] Using an appropriate diagram, explain why the market may supply too little of a merit good.

(b) [15] Using real-world examples, discuss whether every merit good necessarily has a positive consumption externality.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Underappreciated private benefits or positive external benefits can motivate merit-good status; distinguish mechanisms.

Expected diagram or calculation. MSB above MPB, MSC=MPC, Qm<Q*, mark missing social surplus.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Underappreciated private benefits or positive external benefits can motivate merit-good status; distinguish mechanisms.

Counterargument and conditions. Information, capacity and willingness differ; subsidy need not ensure uptake.

Application and real-world example. UK vaccination programs, with documented external benefit mechanism.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 24

Medium | SL/HL | 25 marks

Topic 2.8: Positive consumption

(a) [10] Using an appropriate diagram, explain how a subsidy to consumers could reduce underconsumption of a good with external benefits.

(b) [15] Using real-world examples, evaluate whether lowering price is sufficient to raise uptake.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Subsidy lowers private cost and increases Q toward social optimum if price is barrier.

Expected diagram or calculation. MSB above MPB, MSC=MPC, Qm<Q*, mark missing social surplus.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Subsidy lowers private cost and increases Q toward social optimum if price is barrier.

Counterargument and conditions. Information, capacity and willingness differ; subsidy need not ensure uptake.

Application and real-world example. UK vaccination programs, with documented external benefit mechanism.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 25

Medium | SL/HL | 25 marks

Topic 2.8: Positive consumption

(a) [10] Explain why direct government provision may address underconsumption of a service with positive external benefits.

(b) [15] Using real-world examples, discuss whether provision guarantees the socially optimal level of use.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Free/low-cost access can increase Q but capacity and awareness matter.

Expected diagram or calculation. MSB above MPB, MSC=MPC, Qm<Q*, mark missing social surplus.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes positive consumption from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Free/low-cost access can increase Q but capacity and awareness matter.

Counterargument and conditions. Information, capacity and willingness differ; subsidy need not ensure uptake.

Application and real-world example. UK vaccination programs, with documented external benefit mechanism.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 26

Easy | SL/HL | 25 marks

Topic 2.8: Commons

(a) [10] Explain why a common-pool resource is rivalrous but difficult to exclude users from.

(b) [15] Using real-world examples, discuss whether access restrictions always improve sustainability.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. One user's catch reduces stock for others, while exclusion is difficult.

Expected diagram or calculation. No prescribed standard externality diagram for common-pool classification; if illustrated explain rival use and difficult exclusion.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes commons from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. One user's catch reduces stock for others, while exclusion is difficult.

Counterargument and conditions. Governance, monitoring and local incentives vary.

Application and real-world example. FAO fisheries and community resource management.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 27

Hard | SL/HL | 25 marks

Topic 2.8: Commons

(a) [10] Explain how individual incentives can lead to overuse of a common-pool resource.

(b) [15] Using real-world examples, evaluate whether the tragedy of the commons is inevitable.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Private gain from extraction exceeds private share of collective depletion cost.

Expected diagram or calculation. No prescribed standard externality diagram for common-pool classification; if illustrated explain rival use and difficult exclusion.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes commons from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Private gain from extraction exceeds private share of collective depletion cost.

Counterargument and conditions. Governance, monitoring and local incentives vary.

Application and real-world example. FAO fisheries and community resource management.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 28

Medium | SL/HL | 25 marks

Topic 2.8: Commons

(a) [10] Explain how unsustainable fishing can produce negative externalities.

(b) [15] Using real-world examples, discuss whether limiting total catches is enough to protect fish stocks.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Current catch lowers future stock and imposes costs on others.

Expected diagram or calculation. No prescribed standard externality diagram for common-pool classification; if illustrated explain rival use and difficult exclusion.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes commons from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Current catch lowers future stock and imposes costs on others.

Counterargument and conditions. Governance, monitoring and local incentives vary.

Application and real-world example. FAO fisheries and community resource management.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 29

Hard | SL/HL | 25 marks

Topic 2.8: Commons

(a) [10] Explain how collective self-governance can help manage a common resource.

(b) [15] Using real-world examples, evaluate whether local user agreements can replace government regulation.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Users agree rules, monitoring and sanctions based on shared incentives.

Expected diagram or calculation. No compulsory standard D/S diagram; explain rival use, non-exclusion and rules.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes commons from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Users agree rules, monitoring and sanctions based on shared incentives.

Counterargument and conditions. Trust, enforcement and outsiders may limit.

Application and real-world example. A named community-managed fishery with verified governance.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 30

Medium | SL/HL | 25 marks

Topic 2.8: Commons

(a) [10] Explain why forest use may require cooperation among users even when no single user can exclude others.

(b) [15] Using real-world examples, discuss whether all common resources face identical management problems.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output.

Required reasoning. Shared depletion makes unilateral restraint costly; forest boundaries differ.

Expected diagram or calculation. No compulsory diagram; distinguish common-pool from pure public good.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes commons from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Shared depletion makes unilateral restraint costly; forest boundaries differ.

Counterargument and conditions. Monitoring and ownership regimes vary.

Application and real-world example. A documented community forestry arrangement.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 31

Easy | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Using an appropriate diagram, explain how a Pigouvian tax can address a negative production externality.

(b) [15] Using real-world examples, evaluate whether a carbon tax can achieve the socially efficient output.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Tax equal to marginal external cost aligns MPC plus tax with MSC.

Expected diagram or calculation. Label MSC above MPC, MSB/MPB, Qm/Q*, tax shift and reduced welfare loss.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Tax equal to marginal external cost aligns MPC plus tax with MSC.

Counterargument and conditions. Measurement and responsiveness complicate.

Application and real-world example. World Bank carbon pricing report.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 32

Medium | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain how tradable permits can limit pollution from firms.

(b) [15] Using real-world examples, discuss whether permit markets are more effective than a carbon tax.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. A cap limits total permitted emissions, trade changes allocation.

Expected diagram or calculation. No compulsory externality-policy diagram; describe cap and exchange accurately.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. A cap limits total permitted emissions, trade changes allocation.

Counterargument and conditions. Monitoring, initial allocation and changing conditions matter.

Application and real-world example. World Bank carbon pricing instruments.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 33

Hard | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain how regulation can reduce a negative externality when emissions are difficult to price.

(b) [15] Using real-world examples, evaluate whether a uniform legal standard is suitable for all firms.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Rules constrain emissions/technology directly.

Expected diagram or calculation. If diagram used show Qm>Q* and plausible policy effect, not a fabricated exact tax.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Rules constrain emissions/technology directly.

Counterargument and conditions. Different abatement costs can make uniform rules expensive.

Application and real-world example. A documented air-quality standard.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 34

Medium | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain how education campaigns can change decisions that generate an externality.

(b) [15] Using real-world examples, discuss whether awareness alone corrects market failure.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Information may change perceived benefits/costs and behaviour.

Expected diagram or calculation. Specify whether diagram shifts private demand toward MSB or changes behaviour.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Information may change perceived benefits/costs and behaviour.

Counterargument and conditions. Underlying incentives and constraints may remain.

Application and real-world example. Public vaccination or litter campaign.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 35

Medium | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Using an appropriate diagram, explain how subsidies can correct a positive consumption externality.

(b) [15] Using real-world examples, evaluate whether subsidising vaccination is enough to reach the social optimum.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Subsidy bridges private/social benefit gap and increases Q.

Expected diagram or calculation. MSB above MPB, MPC=MSC, Qm<Q*, subsidy shifts effective private demand.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Subsidy bridges private/social benefit gap and increases Q.

Counterargument and conditions. Supply capacity and hesitancy may limit.

Application and real-world example. UK vaccination coverage context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 36

Easy | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Using an appropriate diagram, explain how a subsidy can respond to a positive production externality.

(b) [15] Using real-world examples, discuss whether subsidising research benefits all firms equally.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Payment aligns MPC with lower MSC, raising Q.

Expected diagram or calculation. MSC below MPC; market and social quantities; subsidy on supply side.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Payment aligns MPC with lower MSC, raising Q.

Counterargument and conditions. Spillover measurement and firm heterogeneity matter.

Application and real-world example. A documented public research grant.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 37

Hard | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain how direct government provision differs from a subsidy in addressing underconsumption.

(b) [15] Using real-world examples, evaluate which approach better improves access to a merit service.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Public producer supplies directly; subsidy lowers private purchase cost.

Expected diagram or calculation. No compulsory diagram; explain institution and price/access channel.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Public producer supplies directly; subsidy lowers private purchase cost.

Counterargument and conditions. Administrative capacity and providers differ.

Application and real-world example. NHS vaccination provision versus a unit subsidy.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 38

Medium | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain why monitoring matters for a policy limiting use of a common-pool resource.

(b) [15] Using real-world examples, discuss whether rules without enforcement can prevent overuse.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Undetected extraction weakens restraints and stock protection.

Expected diagram or calculation. No compulsory diagram; distinguish catch monitoring from production externality tax.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Undetected extraction weakens restraints and stock protection.

Counterargument and conditions. Community norms can complement formal inspection.

Application and real-world example. FAO fisheries management examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 39

Hard | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain why measured external benefits can differ among consumers of the same service.

(b) [15] Using real-world examples, evaluate whether a uniform subsidy achieves the social optimum in every locality.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Spillover depends on coverage and context, so optimal gap varies.

Expected diagram or calculation. A consistent MSB>MPB diagram may show different gaps; avoid exact values without data.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Spillover depends on coverage and context, so optimal gap varies.

Counterargument and conditions. Uniform program may lower administration costs.

Application and real-world example. Vaccination uptake and local transmission conditions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 40

Medium | SL/HL | 25 marks

Topic 2.8: Policies

(a) [10] Explain why a policy correcting one externality might impose costs on other stakeholders.

(b) [15] Using real-world examples, discuss whether welfare improvement requires that every group benefits.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. Responses include Pigouvian taxes, subsidies, regulation, education, permits, direct provision and collective governance.

Required reasoning. Net social gain can coexist with cost to producers or some consumers.

Expected diagram or calculation. Diagram show Qm to Q* and stakeholder effects; do not imply all gains immediate.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes policies from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Net social gain can coexist with cost to producers or some consumers.

Counterargument and conditions. Compensation and equity are separate considerations.

Application and real-world example. Carbon tax on energy-intensive industry.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 41

Easy | SL/HL | 25 marks

Topic 2.8: International

(a) [10] Explain why a global environmental externality may require international cooperation.

(b) [15] Using real-world examples, evaluate whether action by one country is sufficient to address climate change.

Part (a) - Indicative answer and marking guidance

Essential concepts. Cross-border externalities and shared resources require cooperation, monitoring and enforcement; incentives to free-ride make agreement difficult.

Required reasoning. Emissions cross borders and unilateral action leaves shared harm.

Expected diagram or calculation. No compulsory global market diagram; identify cross-border external cost.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes international from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Emissions cross borders and unilateral action leaves shared harm.

Counterargument and conditions. Large country action can still contribute; agreements face free-riding.

Application and real-world example. UNFCCC Paris Agreement.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 42

Medium | SL/HL | 25 marks

Topic 2.8: International

(a) [10] Explain why monitoring is important in an international environmental agreement.

(b) [15] Using real-world examples, discuss whether voluntary commitments can overcome free-riding.

Part (a) - Indicative answer and marking guidance

Essential concepts. Cross-border externalities and shared resources require cooperation, monitoring and enforcement; incentives to free-ride make agreement difficult.

Required reasoning. Transparent reporting helps verify action and reciprocity.

Expected diagram or calculation. No compulsory diagram; distinguish monitoring from actual reductions.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes international from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Transparent reporting helps verify action and reciprocity.

Counterargument and conditions. Reporting alone cannot guarantee enforcement.

Application and real-world example. UNFCCC transparency framework.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 43

Hard | SL/HL | 25 marks

Topic 2.8: International

(a) [10] Explain how the incentive to free-ride can complicate international protection of common resources.

(b) [15] Using real-world examples, evaluate whether cooperation is possible without a supranational authority.

Part (a) - Indicative answer and marking guidance

Essential concepts. Cross-border externalities and shared resources require cooperation, monitoring and enforcement; incentives to free-ride make agreement difficult.

Required reasoning. Each actor benefits from others' restraint while avoiding own cost.

Expected diagram or calculation. No compulsory diagram; identify joint incentives.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes international from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Each actor benefits from others' restraint while avoiding own cost.

Counterargument and conditions. Repeated interaction and shared rules may sustain cooperation.

Application and real-world example. Shared fisheries agreement or Paris climate process.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 44

Medium | SL/HL | 25 marks

Topic 2.8: International

(a) [10] Explain why countries may disagree over how to share the costs of reducing pollution.

(b) [15] Using real-world examples, discuss whether equal reductions by all countries are an equitable solution.

Part (a) - Indicative answer and marking guidance

Essential concepts. Cross-border externalities and shared resources require cooperation, monitoring and enforcement; incentives to free-ride make agreement difficult.

Required reasoning. Different historical emissions, incomes and adjustment costs create disputes.

Expected diagram or calculation. No compulsory diagram; link cost sharing to cooperation.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes international from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Different historical emissions, incomes and adjustment costs create disputes.

Counterargument and conditions. A common target may aid coordination; fairness criteria differ.

Application and real-world example. UNFCCC differentiated contributions.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 45

Medium | SL/HL | 25 marks

Topic 2.8: International

(a) [10] Explain how a carbon tax and an international agreement can complement each other.

(b) [15] Using real-world examples, evaluate whether national carbon pricing is sufficient for global sustainability.

Part (a) - Indicative answer and marking guidance

Essential concepts. Cross-border externalities and shared resources require cooperation, monitoring and enforcement; incentives to free-ride make agreement difficult.

Required reasoning. Domestic price changes incentives while agreement broadens coverage/coordination.

Expected diagram or calculation. Draw negative production externality only for domestic tax mechanism if required.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes international from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Domestic price changes incentives while agreement broadens coverage/coordination.

Counterargument and conditions. Coverage gaps, enforcement and political constraints remain.

Application and real-world example. World Bank carbon pricing and Paris framework.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 46

Easy | HL | 25 marks

Topic 2.8: HL welfare calculation

(a) [10] On an externality diagram, marginal social cost exceeds marginal social benefit by $6 at the market quantity, and the gap narrows linearly to zero at the socially optimal quantity 10 units lower. Calculate welfare loss and explain its area.

(b) [15] Using real-world examples, discuss whether the measured welfare loss captures all environmental harm.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. A linear welfare-loss triangle equals one-half times the quantity gap times marginal social net gap at its widest point.

Required reasoning. Triangle area 0.5*10*6=$30 between Q* and Qm.

Expected diagram or calculation. MSC/MSB gap, market and social Q, triangular area base10 height6.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes hl welfare calculation from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Triangle area 0.5*10*6=$30 between Q* and Qm.

Counterargument and conditions. Valuation and unmeasured harms complicate.

Application and real-world example. Carbon-emission damage; figures hypothetical.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 47

Hard | HL | 25 marks

Topic 2.8: HL welfare calculation

(a) [10] A positive-consumption externality creates a linear marginal-benefit gap of $8 at market quantity that closes at the social optimum 5 units higher. Calculate the welfare loss.

(b) [15] Using real-world examples, evaluate whether a subsidy equal to an estimated external benefit guarantees efficiency.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. A linear welfare-loss triangle equals one-half times the quantity gap times marginal social net gap at its widest point.

Required reasoning. Triangle area .5*5*8=$20, lost net gain from underconsumption.

Expected diagram or calculation. MSB above MPB, Qm<Q*, loss triangle with base5 height8.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes hl welfare calculation from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Triangle area .5*5*8=$20, lost net gain from underconsumption.

Counterargument and conditions. Measurement, changing gap and uptake conditions matter.

Application and real-world example. Vaccination spillover context.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 48

Medium | HL | 25 marks

Topic 2.8: HL welfare calculation

(a) [10] For a negative-consumption externality, excess output is 12 units and the linear excess of MSC over MSB at market output is $4. Calculate the welfare loss.

(b) [15] Using real-world examples, discuss whether restricting consumption to the estimated optimum is straightforward.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. A linear welfare-loss triangle equals one-half times the quantity gap times marginal social net gap at its widest point.

Required reasoning. Area .5*12*4=$24, assuming linear gap closes at Q*; identify negative consumption MSB below MPB.

Expected diagram or calculation. Correct diagram has MSB below MPB, MPC=MSC, Qm>Q*, vertical MSC-MSB gap4 at Qm.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes hl welfare calculation from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Area .5*12*4=$24, assuming linear gap closes at Q*; identify negative consumption MSB below MPB.

Counterargument and conditions. Behavioural response and estimation matter.

Application and real-world example. Second-hand smoke setting, hypothetical numbers.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 49

Hard | HL | 25 marks

Topic 2.8: HL welfare calculation

(a) [10] A positive-production externality has an 8-unit gap between market and social output and a linear marginal net gain of $5 at market output. Calculate welfare loss.

(b) [15] Using real-world examples, evaluate whether research subsidies can remove the entire loss.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. A linear welfare-loss triangle equals one-half times the quantity gap times marginal social net gap at its widest point.

Required reasoning. Triangle .5*8*5=$20 from forgone units.

Expected diagram or calculation. MSC below MPC, market/social Q, underproduction triangle base8 height5.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes hl welfare calculation from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Triangle .5*8*5=$20 from forgone units.

Counterargument and conditions. Spillover variation and imperfect targeting remain.

Application and real-world example. R&D spillover example.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 50

Medium | HL | 25 marks

Topic 2.8: HL welfare calculation

(a) [10] An externality diagram shows market output 40 and efficient output 30, with the marginal social cost exceeding marginal social benefit by $7 at output 40 and a linear gap. Calculate welfare loss and explain why it is not simply 10 times $7.

(b) [15] Using real-world examples, discuss whether accurate welfare-loss estimates are essential before taking action.

Part (a) - Indicative answer and marking guidance

Essential concepts. Social optimum is where MSB=MSC. An externality is a benefit or cost to third parties not fully reflected in private decisions; market output can exceed or fall short of efficient output. A linear welfare-loss triangle equals one-half times the quantity gap times marginal social net gap at its widest point.

Required reasoning. Triangle .5*(40-30)*7=$35, because gap grows from zero at Q30.

Expected diagram or calculation. MSC above MPC, Qm40 Q*30, triangular area; rectangle $70 is incorrect.

Common misconceptions. Do not reverse Qm and Q* for positive/negative externalities, confuse a common-pool resource with a pure public good, or shade a rectangle instead of triangular welfare loss when the gap closes linearly. For this question, check specifically that the student distinguishes hl welfare calculation from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Triangle .5*(40-30)*7=$35, because gap grows from zero at Q30.

Counterargument and conditions. Urgency and uncertainty can justify cautious action.

Application and real-world example. Polluting production context, figures hypothetical.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Sources and scope

The current IB Economics guide controls syllabus boundaries. The uploaded Economics for the IB Diploma, second edition (2012), is a supplementary conceptual check; its older chapter sequence and content cannot expand the current specification. Real-world examples are optional and should be checked against their cited sources.

G. IB Economics guide, first assessment 2022, Topic 2.8, printed pp. 30-31.

Open source

A. World Bank, State and Trends of Carbon Pricing 2024, report and coverage context.

Open source

B. FAO, State of World Fisheries and Aquaculture 2024, fisheries/aquaculture context.

Open source

C. UNFCCC, The Paris Agreement, international climate-cooperation framework.

Open source

Final checks

50 paired questions; 10 Easy / 25 Medium / 15 Hard; Questions 1-45 are SL/HL. Questions 46-50 include HL-only welfare-loss calculations. The student paper contains no answers or difficulty labels.